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Effects of caffeine on P300: A systematic review and meta-analysis

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Abstract

Background: Caffeine is a widely consumed psychoactive substance that stimulates the central nervous system, enhancing alertness and concentration. Its effects on brain activity can be examined using neurophysiological techniques such as event-related potentials, with emphasis on the P300 component. Caffeine appears to modulate P300 amplitude and latency, suggesting changes in processing speed and attention, although findings remain inconsistent. The objective of this work was to systematise the literature about the effects of caffeine on the amplitude and latency of the P300.

Methods: A systematic review and meta-analysis of empirical studies measuring the effect of caffeine on P300 amplitude and latency was conducted, following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses. Statistical analyses were performed using the meta-analysis package for R.

Results: Caffeine intake between 201 and 399 mg was associated with a significant increase in P300 amplitude across anterior, central, and posterior regions, whereas higher doses were associated with smaller effect sizes. No significant differences in P300 latency were observed between pre- and post-caffeine intake.

Discussion: Overall, our findings align with previous reviews showing that caffeine, as a stimulant, can modulate brain activity. The main limitation of this work is the high heterogeneity across studies. Future research directions are discussed.

Keywords: Caffeine; P300; ERP/EEG; Systematic review; Meta-analysis

Journal Pre-proof

Introduction

Caffeine is one of the most widely consumed stimulants worldwide, whether in its pure form or through foods and beverages such as coffee, green or black tea, cocoa, guarana seeds, and soft drinks (Barcelos et al., 2020; Fiani et al., 2021; Irwin et al., 2020; Kuiper, 2022; Soós et al., 2021). Once consumed, it is completely absorbed in the gastrointestinal tract within 30-60 minutes (Soós et al., 2021). After absorption caffeine enters the bloodstream and is distributed through the body before being subsequently metabolized in the liver (Barcelos et al., 2020). Functionally, caffeine acts as a central nervous system (CNS) stimulant, enhancing neuronal activity by blocking adenosine receptors in both the spinal cord and brain (Barcelos et al., 2020; Fiani et al., 2021; Kuiper, 2022; Soós et al., 2021). Adenosine functions as an inhibitory neuromodulator in the CNS, inducing calming effects and promoting sleep when it binds to its receptors (Porkka-Heiskanen, 1999; Soós et al., 2021). By preventing this binding, caffeine induces the opposite effect, leading to increased alertness, energy, concentration, and memory capacity, improved mood, reduced fatigue perception, and ergogenic benefits in physical performance and recovery (Smith, 2002). However, the effects of caffeine may exhibit interindividual variability (Barcelos et al., 2020; Fiani et al., 2021; Kuiper et al., 2022; Soós et al., 2021), with benefits being particularly noticeable under conditions of fatigue and sleep deprivation (Lorist & Tops, 2003; Smith, 2002).

To better understand the effects of caffeine intake, neurophysiological techniques, such as event-related potentials (ERP), are frequently employed (Lorist & Tops, 2003; Veiga et al., 2024). ERPs, derived from an electroencephalogram (EEG), assess neural activity associated with neurocognitive processes such as alertness, attention, and short-term memory (Irwin et al., 2020; Sowndhararajan et al., 2018). Among the various ERP components, the P300 has received particular attention. The P300 is an ERP component

characterized by a positive deflection that occurs approximately 350 to 600 ms after stimulus presentation (Luck, 2014), and is thought to reflect various neurocognitive processes, such as those related to attention and memory (Barry et al., 2019; Polich & Herbst, 2000; Veiga et al., 2004). To elicit the P300, paradigms such as the oddball and the go/no-go tasks are commonly used. The oddball task involves the discrimination - by means of a verbal or motor response - of a rare stimulus in detriment of a frequent one (Polich & Herbst, 2000; Polich, 2007; Sowndhararajan et al., 2018). The go/no-go requires either a motor response (i.e., “go”) or the inhibition of the prepared go response (i.e., “no-go”; Barry et al., 2019; Georgiou & Essau, 2011). During such tasks, greater amplitudes and shorter latencies of the P300 component suggests augmented neurocognitive performance, with faster information processing and improved attention and memory (Kuiper, 2022; Polich & Herbst, 2000; Veiga et al., 2004).

Moreover, the P300 wave encompasses two subcomponents with distinct functional and topographic characteristics (Polich, 2003). The P3a is related to frontal lobe activation, representing attentional orientation and processing of novel or unexpected stimuli (Polich, 2003, 2007). In contrast, the P3b is linked to temporal/parietal functioning that mirrors memory updating and storage after stimulus evaluation (Polich, 2003, 2007). These subcomponents are thought to reflect sequential neuronal activation, with earlier anterior activity being propagated to the central regions (Polich, 2003). Given that caffeine has been consistently shown to improve performance in simple attentional tasks while also potentially enhancing higher-order processes (Einöther & Giesbrecht, 2013), its effects on neurocognitive functioning might be further explored through the P3a and P3b subcomponents.

Prior literature has also touched upon other electrophysiological indices of brain activity following caffeine intake. Modulation of earlier ERP components, such as P200

and N200, seems to support that caffeine enhances early attentional processes and inhibitory control (Chen et al., 2020; Lorist & Tops, 2003). Convergenly, evidence from spectral EEG analysis indicate that caffeine intake is associated with reductions in alpha, delta, and theta frequency (e.g., Barry et al., 2011; Bruce et al., 1986; Chowdhury et al., 2025; Deslandes et al., 2005; Gardiner et al., 2023; Thölke et al., 2025) and increases in beta and sigma bands (e.g., Carrier et al., 2009; Chowdhury et al., 2025; Drapeau et al., 2005; Gardiner et al., 2023). These changes in low- and high-frequency activity point towards heightened cortical arousal and alertness, reduced sleep depth, and higher attentional engagement (e.g., Thölke et al., 2025). Taken together, this evidence suggests that caffeine affects brain activity and neurocognitive functioning across both time- and frequency-domain measures. Considering that P300 is a reliable index of attentional resource allocation, working memory updating and neurocognitive processing (Polich, 2003), it is particularly relevant to examine how caffeine influences this component, since shorter P300 latencies and higher amplitudes are associated with superior mental performance (Sur & Sinha, 2009). Additionally, caffeine's stimulatory effects involve increased activity in noradrenaline neurons and release of dopamine (Nehlig et al., 1992), which may reflect alterations in P3a, thought to be mediated by dopaminergic activity, and P3b, thought to be modulated by the locus coeruleus-norepinephrine system (Huang et al., 2015).

The effects of caffeine have been widely investigated, with previous reviews examining its influence on neurocognitive functioning. Although these reviews provide valuable insights, their findings remain inconclusive. For instance, one review reported predominantly positive effects of caffeine on information processing, reaction time, attention, and memory across several studies (Irwin et al., 2020). While another review, which also focused on neurocognition, concluded that caffeine improves alertness, but its

effects are influenced by multiple factors, including the individual's baseline state (e.g., fatigue, sleep deprivation), dose, and method of ingestion (McLellan et al., 2016). To the best of our knowledge, the most recent systematic review on the effects of caffeine on P300 was published over two decades ago and did not include a meta-analysis (Lorist & Tops, 2003). Moreover, it focused on the neurochemical mechanisms underlying the central effects of caffeine, providing only limited information on the P300 findings. Lorist et al. (2003) also addressed multiple ERP components beyond the P300 (e.g., N1, P2, N2b), resulting in a broader discussion of electrophysiological effects rather than a focused analysis of P300 measures specifically. Still, its results suggest that caffeine acts as a stimulant and can influence several neurocognitive functions, especially in conditions of fatigue and wakefulness (Lorist & Tops, 2003).

Despite the potential of ERP measures to provide precise insights into caffeine modulation of neurocognitive mechanisms, the limited number of available studies has prevented previous quantitative analysis. For this reason, the present work aims to provide a systematic review and meta-analysis of the existing literature regarding the effects of caffeine on P300 amplitude and latency.

Method

Registration

Created on the Open Science Framework on March 16, 2024 and pre-registered on September 29, 2024 (<https://doi.org/10.17605/OSF.IO/5WHCR>), this study adheres to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA; Page et al., 2021) guidelines.

Search strategy

For study retrieval, the databases searched included: PubMed, Web of Science, and EBSCO (PsycArticles, PsycInfo, and Psychology and Behavioral Sciences Collection). The search expression was: (coffe* OR caffein* OR decaff* OR tea OR cola OR chewing gum OR energy drink OR methylxanthin* OR guaran*) AND (ERP OR event related OR evoked potential OR P3) (title/abstract). In addition, studies were also identified by scanning the references from literature reviews and the included studies, and by manual search.

Eligibility criteria

Inclusion criteria were defined by the research team as follows: (a) empirical studies presenting quantitative data; (b) participants aged 18 or above; (c) comparison of P300 amplitude and/or latency before and after caffeine consumption, and/or between the latter and a control group that did not consume caffeine; and (d) studies published in English. Studies were excluded if they were: literature reviews, editorials, commentaries, book chapters, case studies, or qualitative research. Additionally, for the meta-analysis, the articles that did not report quantitative data on ERP amplitude and latency were excluded.

Study selection

Records identified from the different databases were imported into Rayyan (Ouzzani et al., 2016). Duplicates were automatically detected by the software and manually inspected and resolved. The screening process was blindly conducted by BBP and JOP in two stages: initially by title and abstract, followed by a full-text review. Disagreements between both researchers were discussed and settled in a team meeting, during which PM acted as the tiebreaker.

Data extraction and quality assessment

To systematically organize the information to be extracted, a spreadsheet served to capture the following variables: (a) study information – authors, title, and year of publication; (b) sample characteristics – sample size and percentage of female participants, and mean age and standard deviation; (c) caffeine-related information – daily consumption, treatment administered (type and dose), and time between caffeine administration and EEG recording; (d) ERP amplitude and latency data – either the mean and standard deviation, *t*-test and *p*-value, or only the *p*-value, and the effect's direction; (e) ERP characteristics – nomenclature, time window, electrode or electrode cluster, hemisphere, region (anterior, central, or posterior), and type of amplitude measurement (e.g., baseline to peak); and (f) task description – name, modality (auditory or visual), and type of stimulus (e.g., images, sounds).

A meta-analysis was conducted when at least three studies reported P300 measurements following caffeine intake, compared either to a pre-consumption time point or to a non-caffeinated control group, ensuring sufficient data for a reliable quantitative synthesis. Its effect sizes were calculated from the means and standard

deviations of the absolute amplitudes and latencies, the sample size, the p -values, and the direction of the observed effects.

All studies included were evaluated on their overall quality through a checklist for reporting EEG and MEG proposed by Keil et al. (2014). Since the present review focuses on ERP components, some items were not considered, and the assessment concentrated on the remaining key topics: hypotheses, participants, recording characteristics and instruments, timing and stimulus parameters, description of data processing steps, measurement procedures and statistical analyses, and figures. To score the items a binary approach was adopted, where “0” corresponded to absence and “1” to the presence of said item. Therefore, higher quality values reflect greater methodological rigor, in a scale varying between 0 and 28. Two researchers independently evaluated each study, after which they met to reconcile discrepancies.

Analytical strategy and meta-analytic methods

Analyses were conducted using the ‘metafor’ add-on package (version 4.8-0) for meta-analyses (Viechtbauer, 2010) with R (version 4.2.3). The pooled effect sizes for P300 amplitudes and latencies were categorized, using Hedges' g (Hedges, 1981), as small ($g \geq |0.20|$), medium ($g \geq |0.50|$), or large ($g \geq |0.80|$; Cohen, 2013). Confidence intervals, p -values and values for Hedges' g were reported for each analysis, and data was visually summarised through generated forest plots.

The variability between studies (i.e., heterogeneity) was evaluated with Q statistics (Cochran, 1954), which was described using I^2 statistic (Higgins, 2011) at a significance level of 0.1 (Egger et al., 1997). Thresholds for the degree of heterogeneity were as follows: low ($I^2 < 25\%$), medium ($25\% < I^2 \leq 50\%$), or high ($I^2 > 50\%$). Unless variability was low, random-effects models were applied. Publication bias was assessed using Egger's test of intercept bias (Egger et al., 1997).

Moderation analyses were performed in cases where there were a minimum of three studies and heterogeneity was found to be statistically significant. For P300 amplitudes, qualitative data (i.e., topographic region, cognitive process, and stimulus modality) were used in the moderation analysis, whereas quantitative variables (i.e., age, percentage of females, and caffeine dose) were employed in meta-regression models.

Results

Search results

To illustrate the study selection process, a detailed flow diagram based on the PRISMA 2020 guidelines was created (Figure 1; Page et al., 2021). Through the database search, a total of 932 records were identified. After removing duplicates ($n = 338$), nonretrievable studies ($n = 4$), excluding non-eligible studies during the title and abstract screening ($n = 418$), and full-text analysis ($n = 138$), 33 studies remained for inclusion in the review. It is worth noting that, among the 388 studies excluded as out of topic during title and abstract screening, most belonged to adjacent fields such as nutrition, pharmacology, and cellular or molecular research, or to unrelated domains (e.g., Enterprise Resource Planning and environmental/soil sciences). Two additional studies identified through manual search were then added, resulting in a total of 35 studies considered for data extraction. A very high agreement between researchers was achieved regarding the overall inclusion and exclusion decisions ($k = .98$; Landis & Koch, 1977).

<<<INSERT FIGURE 1>>>

Study characteristics

The main characteristics of each study included in the systematic review can be found in Table 1.

<<<INSERT TABLE 1>>>

Participants across samples were aged 18 to 64 years, with the proportion of females varying from 0% ($n = 9$) to 100% ($n = 3$). Habitual daily caffeine intake varied

from less than 36 mg to 600 mg per day, or 1 to 8.6 cups. Although 12 of the studies did not report habitual caffeine consumption, the most common pattern was 1 to 4 cups per day ($n = 6$). In respect to treatment administration, studies varied on their adopted methods, with gelatin capsules being the most frequently used ($n = 11$). Caffeine doses were also diverse and inconsistent, ranging from 5 mg ($n = 2$) to 500 mg ($n = 2$) across studies, with 250 mg being the most commonly administered ($n = 6$). The time between caffeine administration and EEG recording varied across studies, and was most commonly between 30 and 60 minutes ($n = 27$), with 30 minutes being the most common interval ($n = 9$).

The most frequently employed task was the oddball paradigm ($n = 19$) followed by the go/no-go task ($n = 6$). Noteworthy, some studies assess more than one task, or the same task in different conditions (e.g., both visual and auditory testing). Regarding the type of stimulus, most studies used visual stimuli ($n = 26$), in the form of letters ($n = 12$), digits ($n = 3$), or figures (e.g., dots; $n = 10$). Auditory stimuli were used in 11 studies and were presented as tones ($n = 9$), sounds ($n = 12$), or clicks ($n = 1$). Two studies did not report the type of stimulus. Most studies referred to the component as P300 ($n = 28$), whereas a smaller number used P3a and/or P3b ($n = 7$), or P3 ($n = 1$). These were recorded at anterior (i.e., frontal; $n = 2$), central ($n = 4$), or posterior (i.e., parietal or occipital; $n = 8$) topographic regions, or across all three ($n = 25$).

Based on their design, studies were classified into two types: (a) between-groups design studies, comparing participants who ingested caffeine with a non-caffeinated control group ($n = 3$); and (b) pre-post design studies, assessing - the same participants pre- and post-caffeine intake ($n = 33$). Given the small number of between-groups design studies available, a meta-analysis could not be conducted. For the pre-post caffeine

studies (within-subjects), two distinct meta-analyses were performed: P300 amplitude and P300 latency.

Quality assessment

The included studies received a minimum quality evaluation of six ($n = 1$) and a maximum of 25 ($n = 1$), with a total average of 17.41. The most common score, obtained by six studies, was 18 (see Table 1).

Meta-analyses of the pre-post design studies

Corresponding authors of studies published from 2010 onwards were contacted to obtain unreported information required for effect size estimation. Of the three authors contacted, only one provided the requested data.

Study characteristics

The studies measured P300 within a 200 to 800 ms time-window, most frequently being specifically analyzed between 250 and 500 ms ($n = 4$). Due to the considerable variability of the time-windows reported and their unavoidable overlap, the components P300, P3a, and P3b were grouped for the analysis. Additionally, the limited number of studies reporting them separately would reduce statistical power.

Meta-analysis on P300 amplitude

This meta-analysis comprised 28 studies with a combined sample of 523 healthy adults assessed pre- and post-caffeine intake. Overall, a statistically significant difference was found between both time points on P300 amplitude ($g = 0.30$, 95 % CI [0.12, 0.48], $p = .001$, see Figure 2), with the post-caffeine time point revealing an enhancement of

P300 amplitude. Heterogeneity analysis revealed a high level of variance among studies ($Q(27) = 136.86, p < .001, I^2 = 86\%$, see Table 2), but no evidence of publication bias ($b = 0.411, p = .631$).

<<<INSERT FIGURE 2>>>

<<<INSERT TABLE 2>>>

Subgroup analysis

The high heterogeneity detected prompted a moderation analysis to identify potential sources of variability, which are presented in Table 2. When studies were aggregated by topographic regions: anterior (frontal, $k = 5$), central ($k = 9$), posterior (parietal/occipital, $k = 11$), or all ($k = 15$), an amplitude enhancement post-caffeine was found for anterior/central compared to posterior regions ($p < .011$; see Figure 3), probably reflecting P3a (Polich, 2003). No significant effect was found for all regions grouped ($p = .134$). Additionally, although grouping by region reduced heterogeneity, it remained high ($I^2 < 83\%$).

<<<INSERT FIGURE 3>>>

In line with a previous study (Irwin et al., 2020), studies were also grouped by caffeine dose used: ≤ 200 mg ($k = 6$), 201 - 399 mg ($k = 9$), and ≥ 400 mg ($k = 5$). Only the studies that used 201 to 399 mg revealed a statistically significant enhancement of P300 amplitude ($g = 0.33, 95\% \text{ CI } [0.11, 0.55], p = .004$; see Figure 4). No differences were found for the other categorical doses ($ps > .072$), and heterogeneity remained high between studies ($I^2 > 78\%$). Additionally, although not statistically significant, the meta-analytic effect size for studies that used higher doses (≥ 400 mg) revealed a negative effect ($g = -0.17$), indicating that these doses lead to a smaller P300 post-consumption.

Finally, moderation analysis by cognitive processes (attentional vs. executive, Irwin et al., 2020) and by stimulus type (auditory vs. visual) did not lead to a significant reduction in heterogeneity ($I^2 > 80\%$) and reported similar effects ($0.24 < g < 0.40$, $p > .001$).

<<<INSERT FIGURE 4>>>

Meta-regression

Furthermore, we performed a random-effects meta-regression to explore the relationship between the effect sizes of the included studies and the percentage of females, age, and caffeine dose (see Table 3). These analyses indicated that higher caffeine doses were associated with a reduction in effect size, as well as a reversal in direction. Studies administering higher doses tended to report smaller post-caffeine P300 amplitudes ($B = -0.00$, $p = .036$, see Figure 5). No other analysis was statistically significant ($p > .052$).

<<<INSERT TABLE 3>>>

<<<INSERT FIGURE 5>>>

Meta-analysis on P300 latency

This analysis encompassed 15 studies, comprising a total of 269 healthy adults assessed pre- and post-caffeine intake. Overall, we found no statistically significant differences in P300 latencies between both time points ($g = 0.06$, 95% CI [-0.19, 0.30], $p = .654$). The heterogeneity test indicated substantial significant variance across studies ($Q(14) = 54.03$, $p < .001$, $I^2 = 85\%$). There was no evidence of publication bias in this analysis ($b = -0.107$, $p = .876$).

Discussion

This meta-analytical review aimed to summarise the existing literature on the effects of caffeine on the amplitude and latency of the P300 component. The included studies were categorized as either between-groups, comparing participants who ingested caffeine with a non-caffeinated control group, or pre-post caffeine studies, assessing participants pre- and post-caffeine consumption. A meta-analysis of the between-groups studies was not conducted because only three studies were available. For the pre-post caffeine studies, two distinct meta-analyses were conducted: P300 amplitude and P300 latency.

The meta-analysis on P300 amplitude revealed an overall increase in amplitude following caffeine consumption compared to a pre-caffeine time point. This is consistent with previous reviews showing that caffeine, as a stimulant (Lorist & Tops, 2003; Kuiper, 2022), can modulate brain activity (Barry et al., 2007; Kuiper, 2022; Polich & Herbst, 2000). Similar effects on P300 amplitude have been observed with other types of stimulant substances, such as methylphenidate, a synthetic medicine that inhibits neurotransmitters' reuptake, improving neurocognitive functions and brain activity (Barroso et al., 2025).

Due to the high heterogeneity, a moderator analysis was undertaken to explore potential sources of variability, including topographic region, cognitive function, stimuli, and caffeine dose. Grouping studies by stimuli and executive function did not significantly reduce heterogeneity. However, analyses by scalp region showed a modest increase in effect size and reduced heterogeneity for anterior ($g = 0.46$; $I^2 = 67\%$) and central sites ($g = 0.39$; $I^2 = 68\%$), relative to posterior regions ($g = 0.26$; $I^2 = 79\%$). This pattern is broadly consistent with the proposed functional differentiation of P300 subcomponents (P3a vs. P3b), given that anterior recording sites capture more frontal

activity associated with novelty/orienting responses, whereas posterior regions reflect parietal networks involved in context updating (Luck, 2014; Polich, 2003). Thus, it seems as though caffeine exerts a relatively stronger effect over the anterior region, which is likely reflecting modulation of P3a-related attentional orienting mechanisms (Polich, 2003). This is consistent with the notion that caffeine primarily enhances early attentional processes and arousal (McLellan et al., 2016), rather than later-stage working memory processes associated with the P3b component (Polich, 2003). In contrast, studies that collapsed electrodes across regions, such as those using midline sites combined, showed weak, inconsistent effects and maintained high heterogeneity, suggesting that coarse anatomical aggregation obscures subcomponent-specific variability. Taken together, these results support the interpretation that mixing P3a and P3b contributions within a broad 0–800 ms window is likely inflating heterogeneity by combining partly dissociable neural mechanisms.

An interesting moderator, which revealed a non-linear pattern, was caffeine dose. The finding that only doses between 201 and 399 mg resulted in P300 amplitude enhancement suggests that while doses below 200 mg may be insufficient to elicit this effect, those above 400 mg might fail to produce further stimulation and may even produce an opposite outcome. Consistent with this interpretation, the meta-regression analysis showed that higher caffeine doses were associated with reduction in P300 amplitude between pre- and post-caffeine. This result is consistent with a prior study (Ricupero & Ritter, 2024), which found that low doses of caffeine generally yield smaller neurocognitive benefits, whereas higher doses can lead to decreased performance. Together, these findings support the notion of an inverted U-shaped dose–response relationship, in which performance improvements occur up to an optimal caffeine dosage, followed by a subsequent decline.

In the meta-analysis on P300 latency, we found no statistically significant differences in P300 latencies pre- and post-caffeine intake. The small number of studies included in this meta-analysis might have been underpowered to find differences. Nevertheless, this finding aligns with previous research on the effects of caffeine on ERP latency (Souza et al., 2018; Tavanai et al., 2020). P300 peak latency is generally recognised as proportional to stimulus evaluation timing, whereas P300 amplitude reflects the allocation of attentional resources (Polich, 2007). The absence of changes in latency pre- and post-caffeine consumption indicates that neurocognitive processing speed is not substantially altered. Instead, this stimulant's effects might be more likely related to increased arousal, vigilance, learning and mood (McLellan et al., 2016), resulting from its primary role as an antagonist of adenosine A1 and A2 receptors (Institute of Medicine, 2014). Therefore, caffeine might enhance neural activation and attentional resources allocation (Einöther & Giesbrecht, 2013) while not directly accelerating stimulus evaluation speed. Additionally, P300 latency is sensitive to the neurocognitive demands of the employed task and varies according to individual and contextual differences in processing capability (Polich, 2007). Thus, the complexity and familiarity with the task might influence P300 latency, such that processing speed may not be further improved. In fact, caffeine is considered to have more pronounced benefits under conditions that degrade performance, such as fatigue or sleep deprivation (McLellan et al., 2016).

The main limitation of the present work is the high heterogeneity, which may limit the generalizability of the findings. Despite the moderation analysis, the level of heterogeneity did not substantially decrease. This can stem from variability in caffeine administration methods (e.g., capsules, caffeine-containing mixture, pill) or in the tasks and conditions employed to elicit the P300 (e.g. oddball, go/no-go, stroop task), as task

designs can modulate P300 in different ways (Polich & Herbst, 2000; Sugg & Polich, 1995). Another limitation is the focus on electrophysiological measures, overlooking the potential behavioural changes that may accompany them. It is also important to mention that the effects of caffeine on neurocognitive performance depend on the time of consumption and personal sensitivity of the individual (Fiani et al., 2021; Lorist & Tops, 2003; Ricupero & Ritter, 2024; Smith, 2002; Soós et al., 2021). These factors could also be a reason for the observed heterogeneity. Thus, it would be valuable for future studies to integrate, within the same research, the investigation of caffeine's effects on sleep, neurocognitive performance, and brain activity (Lorist & Tops, 2003; Ricupero & Ritter, 2024; Smith, 2002).

Additionally, although the moderator analysis examining the interval between caffeine administration and EEG recording did not reach statistical significance, the result approached the conventional significance threshold ($p = .052$). Given that caffeine absorption and plasma concentrations vary over time following ingestion (Fredholm et al., 1999; Nehlig, 2018), timing-related differences across studies could have potentially contributed, at least to some extent, to variability in the observed findings.

The categorization of caffeine doses into discrete ranges may have limited the sensitivity to detect more nuanced dose–response relationships, as the observed effects may partly reflect the selected categorization strategy rather than a precise physiological threshold. Furthermore, the grouping of P300, P3a, and P3b components may have obscured functionally distinct neurocognitive processes. However, this decision was necessary due to variability and overlap in the time-windows reported across studies.

Considering the results obtained, future studies would benefit from comparing the effects of caffeine on the P300 within the same sample in response to unisensory visual and auditory stimuli, as well as to audiovisual stimuli. Such comparisons would allow for

a more comprehensive understanding of whether caffeine's modulatory effects on neurocognitive processing are specific to modality. Moreover, future research may benefit from disaggregating P300 effects by brain region using multilevel models to better account for within-study variability and increase analytical precision, as well as from distinguishing between P3 subcomponents (e.g., P3a and P3b) to better capture their functionally distinct contributions. P300 has demonstrated clinical value, with associations reported between P300 latency and neurocognitive functioning in various clinical conditions such as neurological disorders, confusional states, and post-traumatic syndrome (Goodin et al., 1983; O'Donnell et al., 1995; Polich & Herbst, 2000). Therefore, studying the effects of caffeine on P300 in clinical populations would be of interest and may inform rehabilitation interventions.

Finally, a better understanding of the effects of caffeine may contribute to evidence-based recommendations for its consumption across diverse settings (e.g., clinical and occupational) and populations (e.g., students, older adults, and clinical groups). More studies are needed to better characterize the dose- and timing-dependent effects of caffeine on EEG measures, in order to improve methodological control and reduce variability across studies.

Altogether, our findings indicate that caffeine primarily modulates arousal-related attentional networks rather than the processing speed of events and stimuli. Furthermore, while moderate doses of caffeine intake (201 to 399 mg) may enhance neurocognitive performance, lower doses might be insufficient to induce a substantial effect and higher doses might provoke overstimulation, thus counteracting the neurocognitive benefits and negatively affecting performance.

Declaration of generative AI and AI-assisted technologies in the writing process

The author(s) did not use generative AI technologies for the preparation of this work. Portions of the R scripts and text editing benefited from AI assistance (ChatGPT, OpenAI). The tool was primarily used to accelerate script development and refine grammar. All analyses, methodological choices, and interpretations were designed, reviewed, and validated by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

* = *Studies that were included in the meta-analysis*

Barcelos, R. P., Lima, F. D., Carvalho, N. R., Bresciani, G., & Royes, L. F. (2020). Caffeine effects on systemic metabolism, oxidative-inflammatory pathways, and exercise performance. *Nutrition Research*, 80, 1–17. <https://doi.org/10.1016/j.nutres.2020.05.005>

Barroso, B. C. R., Mendonça, F., Mazer, P., Prata, C., & Pinto, J. O. (2025). Methylphenidate and P300 in attention deficit hyperactivity disorder: A systematic review and meta-analysis. *International Journal of Psychophysiology*, 211, 112553. <https://doi.org/10.1016/j.ijpsycho.2025.112553>

Barry, R. J., Johnstone, S. J., Clarke, A. R., Rushby, J. A., Brown, C. R., & McKenzie, D. N. (2007). Caffeine effects on ERPs and performance in an auditory Go/NoGo task. *Clinical Neurophysiology : Official Journal of the International Federation of*

Clinical Neurophysiology, 118(12), 2692–2699.
<https://doi.org/10.1016/j.clinph.2007.08.023>*

Barry, R. J., Clarke, A. R., & Johnstone, S. J. (2011). Caffeine and opening the eyes have additive effects on resting arousal measures. *Clinical Neurophysiology*, 122(10), 2010–2015. <https://doi.org/10.1016/j.clinph.2011.02.036>

Barry, R. J., De Blasio, F. M., & Cave, A. E. (2014). Caffeine Effects on ERP Components and Performance in an Equiprobable Auditory Go/NoGo Task. *Journal of Caffeine Research*, 4(3), 83–92. <https://doi.org/10.1089/jcr.2014.0011>*

Barry, R. J., De Blasio, F. M., & Fogarty, J. S. (2019). Caffeine affects children's ERPs and performance in an equiprobable go/no-go task: Testing a processing schema. *Psychophysiology*, 56(5), e13330. <https://doi.org/10.1111/psyp.13330>

Barry, R. J., Fogarty, J. S., & De Blasio, F. M. (2020). Caffeine as a Tool to Explore Active Cognitive Processing Stages in Two-Choice Tasks. *Journal of Caffeine and Adenosine Research*, 10(2), 71–83. <https://doi.org/10.1089/caff.2019.0021>*

Blanco-Díaz, C. F., Guerrero-Méndez, C. D., Bastos-Filho, T., Jaramillo-Isaza, S., & Ruiz-Olaya, A. F. (2022). Effects of the concentration level, eye fatigue and coffee consumption on the performance of a BCI system based on visual ERP-P300. *Journal of Neuroscience Methods*, 382, 109722. <https://doi.org/10.1016/j.jneumeth.2022.109722>

Bruce, M., Scott, N., Lader, M., & Marks, V. (1986). The psychopharmacological and electrophysiological effects of single doses of caffeine in healthy human subjects. *British Journal of Clinical Pharmacology*, 22(1), 81–87. <https://doi.org/10.1111/j.1365-2125.1986.tb02883.x>

Carrier, J., Paquet, J., Fernandez-Bolanos, M., Girouard, L., Roy, J., Selmaoui, B., & Filipini, D. (2009). Effects of caffeine on daytime recovery sleep: A double challenge to the sleep–wake cycle in aging. *Sleep Medicine*, 10(9), 1016–1024. <https://doi.org/10.1016/j.sleep.2009.01.001>

Chen, X., Zhang, L., Yang, D., Li, C., An, G., Wang, J., Shao, Y., Fan, R., & Ma, Q. (2020). Effects of Caffeine on Event-Related Potentials and Neuropsychological

Indices After Sleep Deprivation. *Frontiers in Behavioral Neuroscience*, 14, 108.
<https://doi.org/10.3389/fnbeh.2020.00108>

Chowdhury, S., Alanazi, A. M., & Attar, E. T. (2025). Caffeine on the mind: EEG and cardiovascular signatures of cortical arousal revealed by wearable sensors and machine learning—a pilot study on a male group. *Frontiers in Systems Neuroscience*, 19, 1611293. <https://doi.org/10.3389/fnsys.2025.1611293>

Cochran, W. G. (1954). The Combination of Estimates from Different Experiments. *Biometrics*, 10(1), 101–129. <https://doi.org/10.2307/3001666>

Cohen, J. (2013). *Statistical power analysis for the behavioral sciences*. Routledge.
<https://doi.org/10.4324/9780203771587>

Cropley, V., Croft, R., Silber, B., Neale, C., Scholey, A., Stough, C., & Schmitt, J. (2012). Does coffee enriched with chlorogenic acids improve mood and cognition after acute administration in healthy elderly? A pilot study. *Psychopharmacology*, 219(3), 737–749. <https://doi.org/10.1007/s00213-011-2395-0>*

Debrah, K., Sherwin, R. S., Murphy, J., & Kerr, D. (1996). Effect of caffeine on recognition of and physiological responses to hypoglycaemia in insulin-dependent diabetes. *Lancet*, 347(8993), 19–24. [https://doi.org/10.1016/s0140-6736\(96\)91557-3](https://doi.org/10.1016/s0140-6736(96)91557-3)

De Pauw, K., Roelands, B., Knaepen, K., Polfliet, M., Stiens, J., & Meeusen, R. (2015). Effects of caffeine and maltodextrin mouth rinsing on P300, brain imaging, and cognitive performance. *Journal of Applied Physiology*, 118(6), 776–782. <https://doi.org/10.1152/jappphysiol.01050.2014>

De Pauw, K., Roelands, B., Van Cutsem, J., Marusic, U., Torbeyns, T., & Meeusen, R. (2017). Electro-physiological changes in the brain induced by caffeine or glucose nasal spray. *Psychopharmacology*, 234(1), 53–62. <https://doi.org/10.1007/s00213-016-4435-2>*

Deslandes, A. C., Veiga, H., Cagy, M., Piedade, R., Pompeu, F., & Ribeiro, P. (2004). Effects of caffeine on visual evoked potential (P300) and neuromotor performance.

Arquivos de Neuro-psiquiatria, 62(2B), 385–390. <https://doi.org/10.1590/s0004-282x2004000300002>*

Deslandes, A. C., Veiga, H., Cagy, M., Piedade, R., Pompeu, F., & Ribeiro, P. (2005). Effects of caffeine on the electrophysiological, cognitive and motor responses of the central nervous system. *Brazilian Journal of Medical and Biological Research = Revista brasileira de pesquisas medicas e biologicas*, 38(7), 1077–1086. <https://doi.org/10.1590/s0100-879x2005000700011>*

Deslandes, A., Ferreira, C., Veiga, H., Cagy, M., Piedade, R., Pompeu, F., & Ribeiro, P. (2006). Effects of caffeine on electrophysiological and neuropsychological indices after sleep deprivation. *Neuropsychobiology*, 54(2), 126–133. <https://doi.org/10.1159/000098263>*

Dimpfel, W., Kler, A., Kriesl, E., Lehnfeld, R., & Keplinger-Dimpfel, I. K. (2007). Source density analysis of the human EEG after ingestion of a drink containing decaffeinated extract of green tea enriched with L-theanine and theogallin. *Nutritional Neuroscience*, 10(3-4), 169–180. <https://doi.org/10.1080/03093640701580610>*

Diukova, A., Ware, J., Smith, J. E., Evans, C. J., Murphy, K., Rogers, P. J., & Wise, R. G. (2012). Separating neural and vascular effects of caffeine using simultaneous EEG-fMRI: differential effects of caffeine on cognitive and sensorimotor brain responses. *NeuroImage*, 62(1), 239–249. <https://doi.org/10.1016/j.neuroimage.2012.04.041>*

Dixit, A., Vaney, N., & Tandon, O. P. (2006). Evaluation of cognitive brain functions in caffeine users: a P3 evoked potential study. *Indian Journal of Physiology and Pharmacology*, 50(2), 175–180*

Drapeau, C., Hamel-hébert, I., Robillard, R., Selmaoui, B., Filipini, D., & Carrier, J. (2006). Challenging sleep in aging: The effects of 200 mg of caffeine during the evening in young and middle-aged moderate caffeine consumers. *Journal of Sleep Research*, 15(2), 133–141. <https://doi.org/10.1111/j.1365-2869.2006.00518.x>

- Egger, M., Davey Smith, G., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *BMJ (Clinical research ed.)*, 315(7109), 629–634. <https://doi.org/10.1136/bmj.315.7109.629>
- Einöther, S. J. L., & Giesbrecht, T. (2012). Caffeine as an attention enhancer: Reviewing existing assumptions. *Psychopharmacology*, 225(2), 251–274. <https://doi.org/10.1007/s00213-012-2917-4>
- Fiani, B., Zhu, L., Musch, B. L., Briceno, S., Andel, R., Sadeq, N., & Ansari, A. Z. (2021). The Neurophysiology of Caffeine as a Central Nervous System Stimulant and the Resultant Effects on Cognitive Function. *Cureus*, 13(5), e15032. <https://doi.org/10.7759/cureus.15032>
- Fredholm, B. B., Bättig, K., Holmén, J., Nehlig, A., & Zvartau, E. E. (1999). Actions of caffeine in the brain with special reference to factors that contribute to its widespread use. *Pharmacological Reviews*, 51(1), 83–133.
- Gardiner, C., Weakley, J., Burke, L. M., Roach, G. D., Sargent, C., Maniar, N., Townshend, A., & Halson, S. L. (2023). The effect of caffeine on subsequent sleep: A systematic review and meta-analysis. *Sleep Medicine Reviews*, 69, 101764. <https://doi.org/10.1016/j.smr.2023.101764>
- Georgiou, G., & Essau, C. A. (2011). *Go/No-Go Task*. In S. Goldstein & J. A. Naglieri (Eds.), *Encyclopedia of Child Behavior and Development* (pp. 705–706). Springer. https://doi.org/10.1007/978-0-387-79061-9_1267
- Goodin, D. S., Starr, A., Chippendale, T., & Squires, K. C. (1983). Sequential changes in the P3 component of the auditory evoked potential in confusional states and dementing illnesses. *Neurology*, 33(9), 1215–1218. <https://doi.org/10.1212/wnl.33.9.1215>
- Hedges, L. V. (1981). Distribution theory for Glass's estimator of effect size and related estimators. *Journal of Educational Statistics*, 6(2), 107–128. <https://doi.org/10.2307/1164588>
- Higgins, J. P., Altman, D. G., Gøtzsche, P. C., Jüni, P., Moher, D., Oxman, A. D., Savovic, J., Schulz, K. F., Weeks, L., Sterne, J. A., Cochrane Bias Methods Group,

- & Cochrane Statistical Methods Group (2011). The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ (Clinical research ed.)*, 343, d5928. <https://doi.org/10.1136/bmj.d5928>
- Hirvonen, J., Jääskeläinen, I. P., Näätänen, R., & Sillanaukee, P. (2000). Adenosine A1/A2a receptors mediate suppression of mismatch negativity by ethanol in humans. *Neuroscience Letters*, 278(1-2), 57–60. [https://doi.org/10.1016/s0304-3940\(99\)00897-6](https://doi.org/10.1016/s0304-3940(99)00897-6)*
- Huang, W., Chen, W., & Zhang, X. (2015). The neurophysiology of P300 - An integrated review. *European Review for Medical and Pharmacological Sciences*, 19(8), 1480–1488.
- Institute of Medicine. (2014). *Caffeine in food and dietary supplements: Examining safety*. National Academies Press. <https://doi.org/10.17226/18607>
- Irwin, C., Khalesi, S., Desbrow, B., & McCartney, D. (2020). Effects of acute caffeine consumption following sleep loss on cognitive, physical, occupational and driving performance: A systematic review and meta-analysis. *Neuroscience and Biobehavioral Reviews*, 108, 877–888. <https://doi.org/10.1016/j.neubiorev.2019.12.008>
- Kahathuduwa, C. N., Dassanayake, T. L., Amarakoon, A. M. T., & Weerasinghe, V. S. (2017). Acute effects of theanine, caffeine and theanine-caffeine combination on attention. *Nutritional Neuroscience*, 20(6), 369–377. <https://doi.org/10.1080/1028415X.2016.1144845>*
- Kawamura, N., Maeda, H., Nakamura, J., Morita, K., & Nakazawa, Y. (1996). Effects of caffeine on event-related potentials: comparison of oddball with single-tone paradigms. *Psychiatry and Clinical Neurosciences*, 50(4), 217–221. <https://doi.org/10.1111/j.1440-1819.1996.tb02745.x>*
- Kenemans, J. L., & Lorist, M. M. (1995). Caffeine and selective visual processing. *Pharmacology, biochemistry, and behavior*, 52(3), 461–471. [https://doi.org/10.1016/0091-3057\(95\)00159-t](https://doi.org/10.1016/0091-3057(95)00159-t)

- Kerr, D., Sherwin, R. S., Pavalkis, F., Fayad, P. B., Sikorski, L., Rife, F., Tamborlane, W. V., & During, M. J. (1993). Effect of caffeine on the recognition of and responses to hypoglycemia in humans. *Annals of internal medicine*, 119(8), 799–804. <https://doi.org/10.7326/0003-4819-119-8-199310150-00005>
- Keil, A., Debener, S., Gratton, G., Junghöfer, M., Kappenman, E. S., Luck, S. J., Luu, P., Miller, G. A., & Yee, C. M. (2014). Committee report: publication guidelines and recommendations for studies using electroencephalography and magnetoencephalography. *Psychophysiology*, 51(1), 1–21. <https://doi.org/10.1111/psyp.12147>
- Kuiper, M. (2022). *The effect of habitual caffeine consumption on inhibition* (Bachelor's thesis, Tilburg University).
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>.
- Lorist, M. M., Snel, J., & Kok, A. (1994). Influence of caffeine on information processing stages in well rested and fatigued subjects. *Psychopharmacology*, 113(3-4), 411–421. <https://doi.org/10.1007/BF02245217>*
- Lorist, M. M., Snel, J., Mulder, G., & Kok, A. (1995). Aging, caffeine, and information processing: an event-related potential analysis. *Electroencephalography and Clinical Neurophysiology*, 96(5), 453–467. [https://doi.org/10.1016/0168-5597\(95\)00069-5](https://doi.org/10.1016/0168-5597(95)00069-5)*
- Lorist, M. M., Snel, J., Kok, A., & Mulder, G. (1996). Acute effects of caffeine on selective attention and visual search processes. *Psychophysiology*, 33(4), 354–361. <https://doi.org/10.1111/j.1469-8986.1996.tb01059.x>
- Lorist, M. M., & Snel, J. (1997). Caffeine effects on perceptual and motor processes. *Electroencephalography and Clinical Neurophysiology*, 102(5), 401–413. [https://doi.org/10.1016/s0921-884x\(97\)95729-5](https://doi.org/10.1016/s0921-884x(97)95729-5)*

- Lorist, M. M., & Tops, M. (2003). Caffeine, fatigue, and cognition. *Brain and Cognition*, 53(1), 82–94. [https://doi.org/10.1016/s0278-2626\(03\)00206-9](https://doi.org/10.1016/s0278-2626(03)00206-9)
- Luck, S. J. (2014). *An Introduction to the Event-Related Potential Technique, second edition*. MIT Press.
- Maciejewska, K., & Grabowska, K. (2020). Acute effect of energy boost dietary supplement on P3 waveform: double blind, placebo controlled study. *Acta Neurobiologiae Experimentalis*, 80(4), 411–423*
- Martin, F. H., & Garfield, J. (2006). Combined effects of alcohol and caffeine on the late components of the event-related potential and on reaction time. *Biological Psychology*, 71(1), 63–73. <https://doi.org/10.1016/j.biopsycho.2005.01.004>*
- McLellan, T. M., Caldwell, J. A., & Lieberman, H. R. (2016). A review of caffeine's effects on cognitive, physical and occupational performance. *Neuroscience and Biobehavioral Reviews*, 71, 294–312. <https://doi.org/10.1016/j.neubiorev.2016.09.001>
- Montenegro, M., Veiga, H., Deslandes, A., Cagy, M., McDowell, K., Pompeu, F., Piedade, R., & Ribeiro, P. (2005). Neuromodulatory effects of caffeine and bromazepam on visual event-related potential (P300): a comparative study. *Arquivos de Neuro-psiquiatria*, 63(2B), 410–415. <https://doi.org/10.1590/s0004-282x2005000300009>*
- Nehlig, A., Daval, J., & Debry, G. (1992). Caffeine and the central nervous system: Mechanisms of action, biochemical, metabolic and psychostimulant effects. *Brain Research Reviews*, 17(2), 139–170. [https://doi.org/10.1016/0165-0173\(92\)90012-b](https://doi.org/10.1016/0165-0173(92)90012-b)
- Nehlig A. (2018). Interindividual Differences in Caffeine Metabolism and Factors Driving Caffeine Consumption. *Pharmacological Reviews*, 70(2), 384–411. <https://doi.org/10.1124/pr.117.014407>
- O'Donnell, B. F., Faux, S. F., McCarley, R. W., Kimble, M. O., Salisbury, D. F., Nestor, P. G., Kikinis, R., Jolesz, F. A., & Shenton, M. E. (1995). Increased rate of P300 latency prolongation with age in schizophrenia. *Electrophysiological*

- evidence for a neurodegenerative process. *Archives of General Psychiatry*, 52(7), 544–549. <https://doi.org/10.1001/archpsyc.1995.03950190026004>
- Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan: A web and mobile app for systematic reviews. *Systematic Reviews*, 5, 210. <https://doi.org/10.1186/s13643-016-0384-4>
- Pan, J., Takeshita, T., & Morimoto, K. (2000). Acute caffeine effect on repeatedly measured P300. *Environmental Health and Preventive Medicine*, 5(1), 13–17. <https://doi.org/10.1007/BF02935910>*
- Rao, A., Hu, H., & Nobre, A. C. (2005). The effects of combined caffeine and glucose drinks on attention in the human brain. *Nutritional neuroscience*, 8(3), 141–153. <https://doi.org/10.1080/10284150500096994>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L. A., ... Moher, D. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ (Clinical research ed.)*, 372, n71. <https://doi.org/10.1136/bmj.n71>
- Polich, J. (2003). Theoretical overview of P3a and P3b. In J. Polich (Ed.), *Detection of change* (pp. 83–98). Springer. https://doi.org/10.1007/978-1-4615-0294-4_5
- Polich, J. (2007). Updating P300: An integrative theory of P3a and P3b. *Clinical Neurophysiology*, 118(10), 2128–2148. <https://doi.org/10.1016/j.clinph.2007.04.019>
- Polich, J., & Herbst, K. L. (2000). P300 as a clinical assay: rationale, evaluation, and findings. *International Journal of Psychophysiology : Official Journal of the International Organization of Psychophysiology*, 38(1), 3–19. [https://doi.org/10.1016/s0167-8760\(00\)00127-6](https://doi.org/10.1016/s0167-8760(00)00127-6)
- Porkka-Heiskanen T. (1999). Adenosine in sleep and wakefulness. *Annals of Medicine*, 31(2), 125–129. <https://doi.org/10.3109/07853899908998788>

- Reeves, R. R., Struve, F. A., & Patrick, G. (1999). The effects of caffeine withdrawal on cognitive P300 auditory and visual evoked potentials. *Clinical EEG (electroencephalography)*, 30(1), 24–27. <https://doi.org/10.1177/155005949903000110>*
- Ricupero, S., & Ritter, F. E. (2024). Caffeine and cognition: A cognitive architecture-based review. *Theoretical Issues in Ergonomics Science*, 25(6), 655–679. <https://doi.org/10.1080/1463922X.2024.2323547>
- Ruijter, J., Lorist, M. M., & Snel, J. (1999). The influence of different doses of caffeine on visual task performance. *Journal of Psychophysiology*, 13(1), 37–48. <https://doi.org/10.1027/0269-8803.13.1.37>*
- Ruijter, J., De Ruiter, M. B., & Snel, J. (2000). The effects of caffeine on visual selective attention to color: an ERP study. *Psychophysiology*, 37(4), 427–439 *
- Ruijter, J., Lorist, M. M., Snel, J., & De Ruiter, M. B. (2000). The influence of caffeine on sustained attention: an ERP study. *Pharmacology, Biochemistry, and Behavior*, 66(1), 29–37. [https://doi.org/10.1016/s0091-3057\(00\)00229-x](https://doi.org/10.1016/s0091-3057(00)00229-x)*
- Saville, C. W. N., de Morree, H. M., Dundon, N. M., Marcora, S. M., & Klein, C. (2018). Effects of caffeine on reaction time are mediated by attentional rather than motor processes. *Psychopharmacology*, 235(3), 749–759. <https://doi.org/10.1007/s00213-017-4790-7>*
- Sela I. (2014). Visual and auditory synchronization deficits among dyslexic readers as compared to non-impaired readers: a cross-correlation algorithm analysis. *Frontiers in Human Neuroscience*, 8, 364. <https://doi.org/10.3389/fnhum.2014.00364>
- Smith A. (2002). Effects of caffeine on human behavior. *Food and chemical toxicology : an international journal published for the British Industrial Biological Research Association*, 40(9), 1243–1255. [https://doi.org/10.1016/s0278-6915\(02\)00096-0](https://doi.org/10.1016/s0278-6915(02)00096-0)
- Soós, R., Gyebrovski, Á., Tóth, Á., Jeges, S., & Wilhelm, M. (2021). Effects of Caffeine and Caffeinated Beverages in Children, Adolescents and Young Adults:

Short Review. *International Journal of Environmental Research and Public Health*, 18(23), 12389. <https://doi.org/10.3390/ijerph182312389>

Souza, M. E. D. C. A., Costa, K. V. T. D., & Menezes, P. L. (2018). Effect of caffeine on vestibular evoked myogenic potential: a systematic review with meta-analysis. *Brazilian Journal of Otorhinolaryngology*, 84(3), 381–388. <https://doi.org/10.1016/j.bjorl.2017.11.003>

Sowndhararajan, K., Kim, M., Deepa, P., Park, S. J., & Kim, S. (2018). Application of the P300 Event-Related Potential in the Diagnosis of Epilepsy Disorder: A Review. *Scientia Pharmaceutica*, 86(2), 10. <https://doi.org/10.3390/scipharm86020010>

Sugg, M. J., & Polich, J. (1995). P300 from auditory stimuli: intensity and frequency effects. *Biological Psychology*, 41(3), 255–269. [https://doi.org/10.1016/0301-0511\(95\)05136-8](https://doi.org/10.1016/0301-0511(95)05136-8)

Sur, S., & Sinha, V. (2009). Event-related potential: An overview. *Industrial Psychiatry Journal*, 18(1), 70–73. <https://doi.org/10.4103/0972-6748.57865>

Tavanai, E., Farahani, S., Ghahraman, M. A., Soleimanian, S., & Jalaie, S. (2020). Effects of Caffeine on Auditory- and Vestibular-Evoked Potentials in Healthy Individuals: A Double-Blind Placebo-Controlled Study. *Journal of Audiology & Otology*, 24(1), 10–16. <https://doi.org/10.7874/jao.2019.00227>

Thölke, P., Arcand-Lavigne, M., Lajnef, T., Frenette, S., Carrier, J., & Jerbi, K. (2025). Caffeine induces age-dependent increases in brain complexity and criticality during sleep. *Communications Biology*, 8(1), 685. <https://doi.org/10.1038/s42003-025-08090-z>

Tieges, Z., Snel, J., Kok, A., Plat, N., & Ridderinkhof, R. (2007). Effects of caffeine on anticipatory control processes: evidence from a cued task-switch paradigm. *Psychophysiology*, 44(4), 561–578. <https://doi.org/10.1111/j.1469-8986.2007.00534.x>

Tieges, Z., Richard Ridderinkhof, K., Snel, J., & Kok, A. (2004). Caffeine strengthens action monitoring: evidence from the error-related negativity. *Brain research*.

Cognitive Brain Research, 21(1), 87–93.
<https://doi.org/10.1016/j.cogbrainres.2004.06.001>*

Trunk, A., Stefanics, G., Zentai, N., Bacskey, I., Felinger, A., Thuróczy, G., & Hernádi, I. (2014). Lack of interaction between concurrent caffeine and mobile phone exposure on visual target detection: an ERP study. *Pharmacology, Biochemistry, and Behavior*, 124, 412–420.
<https://doi.org/10.1016/j.pbb.2014.07.011>*

Viechtbauer, W. (2010). Conducting Meta-Analyses in R with the metafor Package. *Journal of Statistical Software*, 36(3), 1–48. <https://doi.org/10.18637/jss.v036.i03>

Vorkapić, T. S. (2009). Biological correlates of P300 and extraversion relationship. *Collegium Antropologicum*, 33(4), 1215–1222.

Veiga, H., Deslandes, A., Cagy, M., McDowell, K., Pompeu, F., Piedade, R., & Ribeiro, P. (2004). Visual event-related potential (P300): a normative study. *Arquivos de Neuro-psiquiatria*, 62(3A), 575–581. <https://doi.org/10.1590/s0004-282x2004000400002>

Wicht, C. A., De Pretto, M., Mouthon, M., & Spierer, L. (2022). Neural correlates of expectations-induced effects of caffeine intake on executive functions. *Cortex*, 150, 61–84. <https://doi.org/10.1016/j.cortex.2022.02.004>*

Xu, L., Yang, X., Peng, Z., Song, T., Wang, L., Dai, C., Xu, M., Shao, Y., & Lv, J. (2022). Modafinil ameliorates the decline in pronunciation-related working memory caused by 36-h acute total sleep deprivation: An ERP study. *Neurobiology of Learning and Memory*, 192, 107625.
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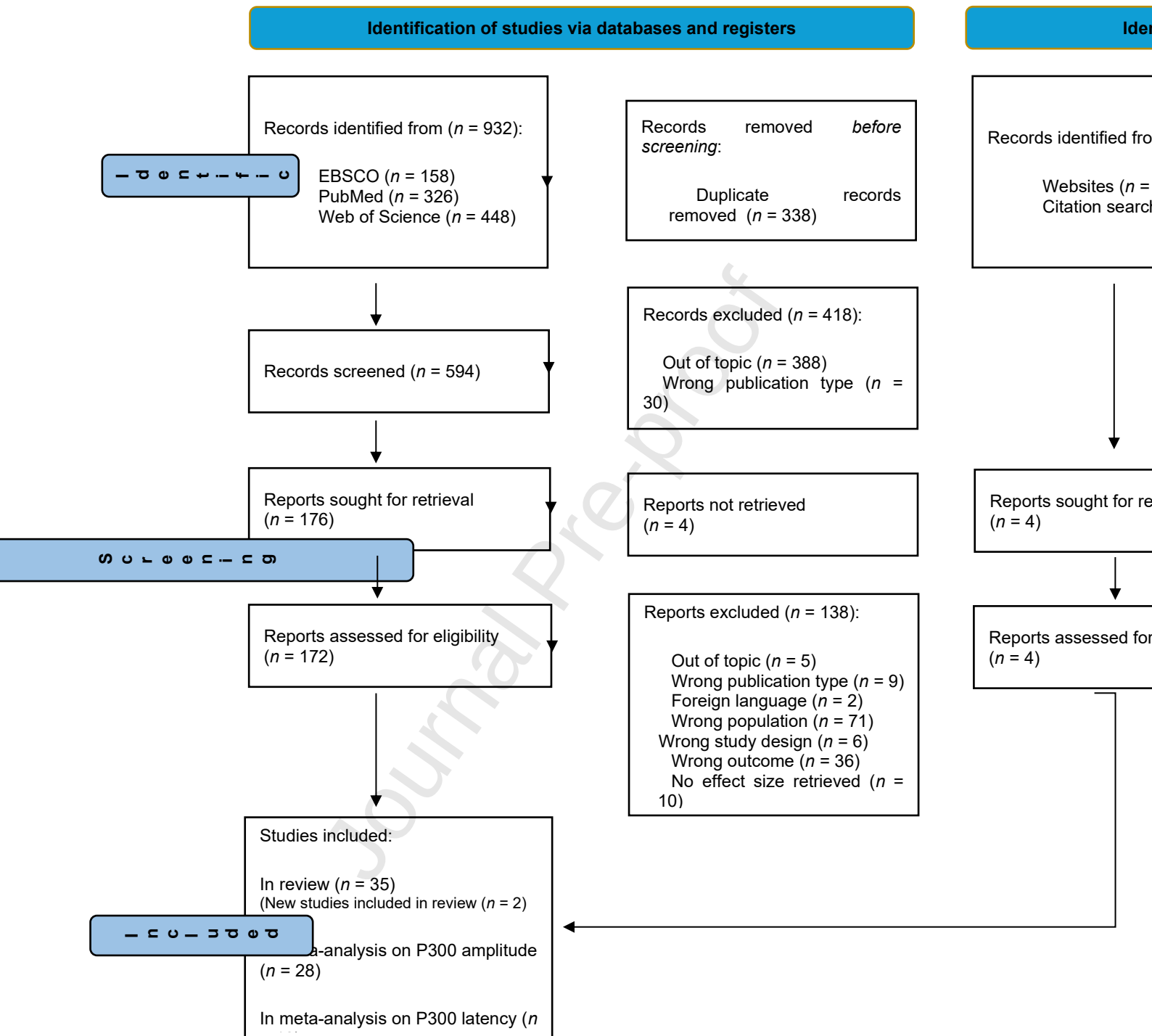
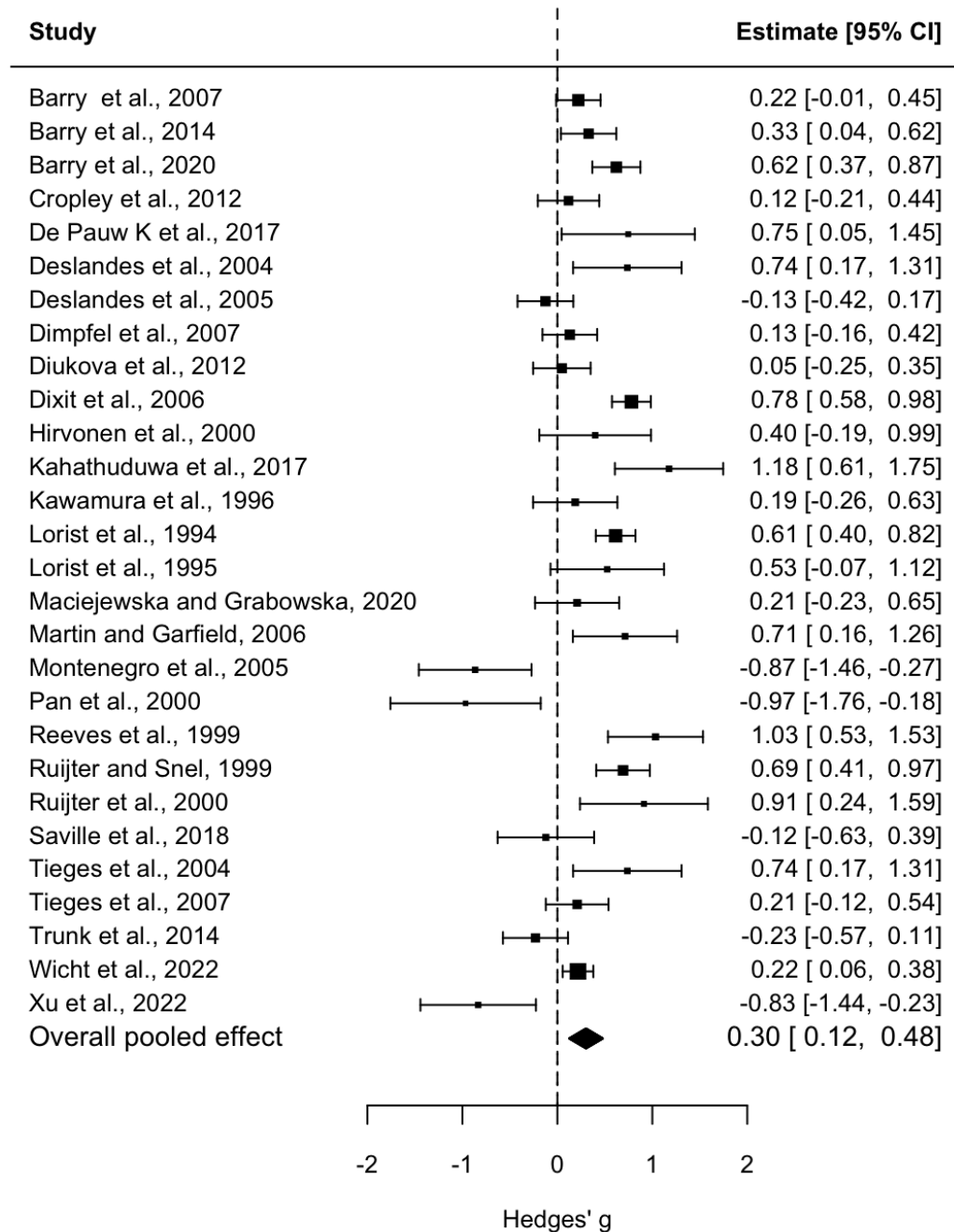
Figure 1*Prisma flow diagram*

Figure 2*Forest plot: Overall effect sizes per study***Figure 3**

Forest plot by region

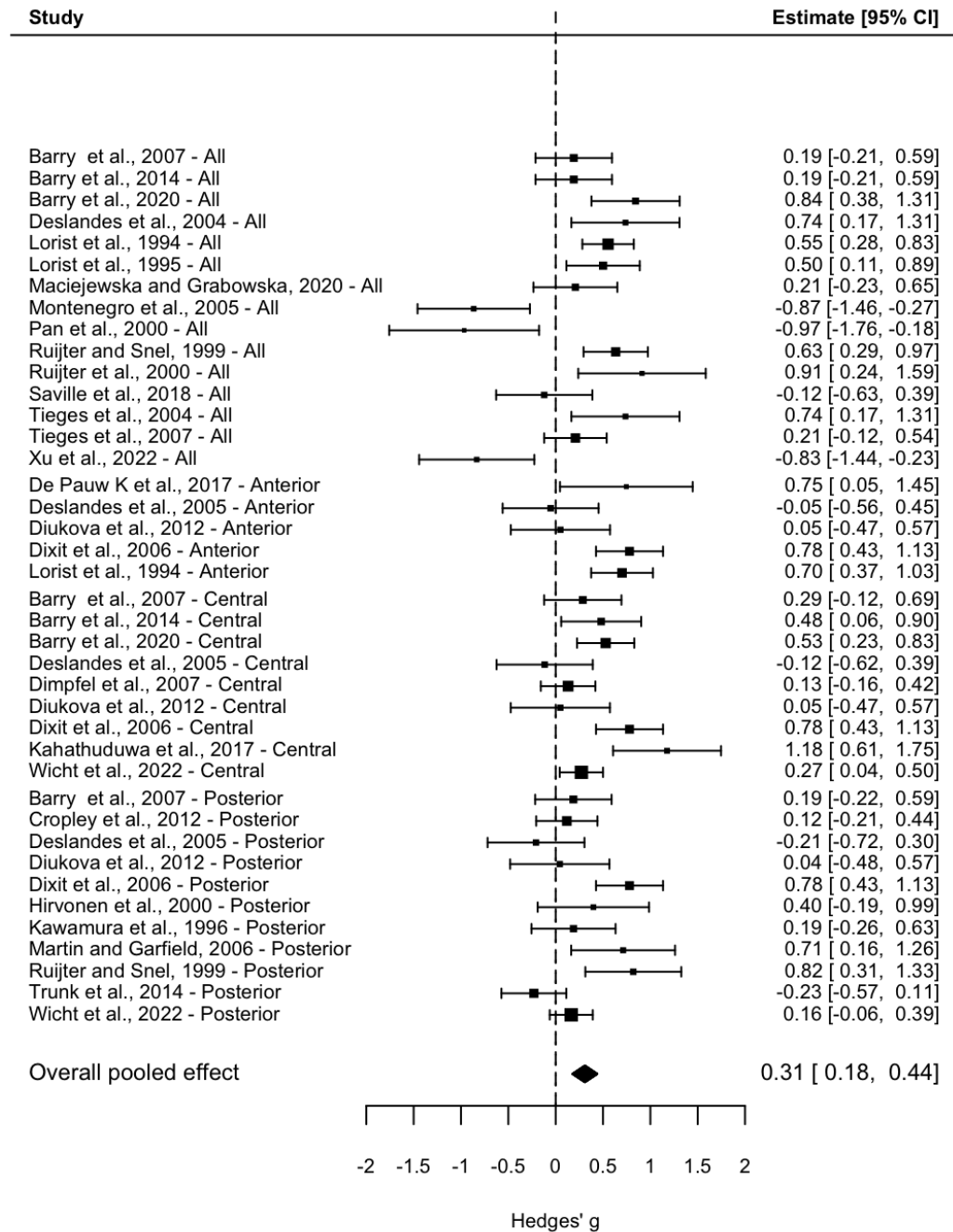


Figure 4

Combined forest plot by categorical caffeine dose

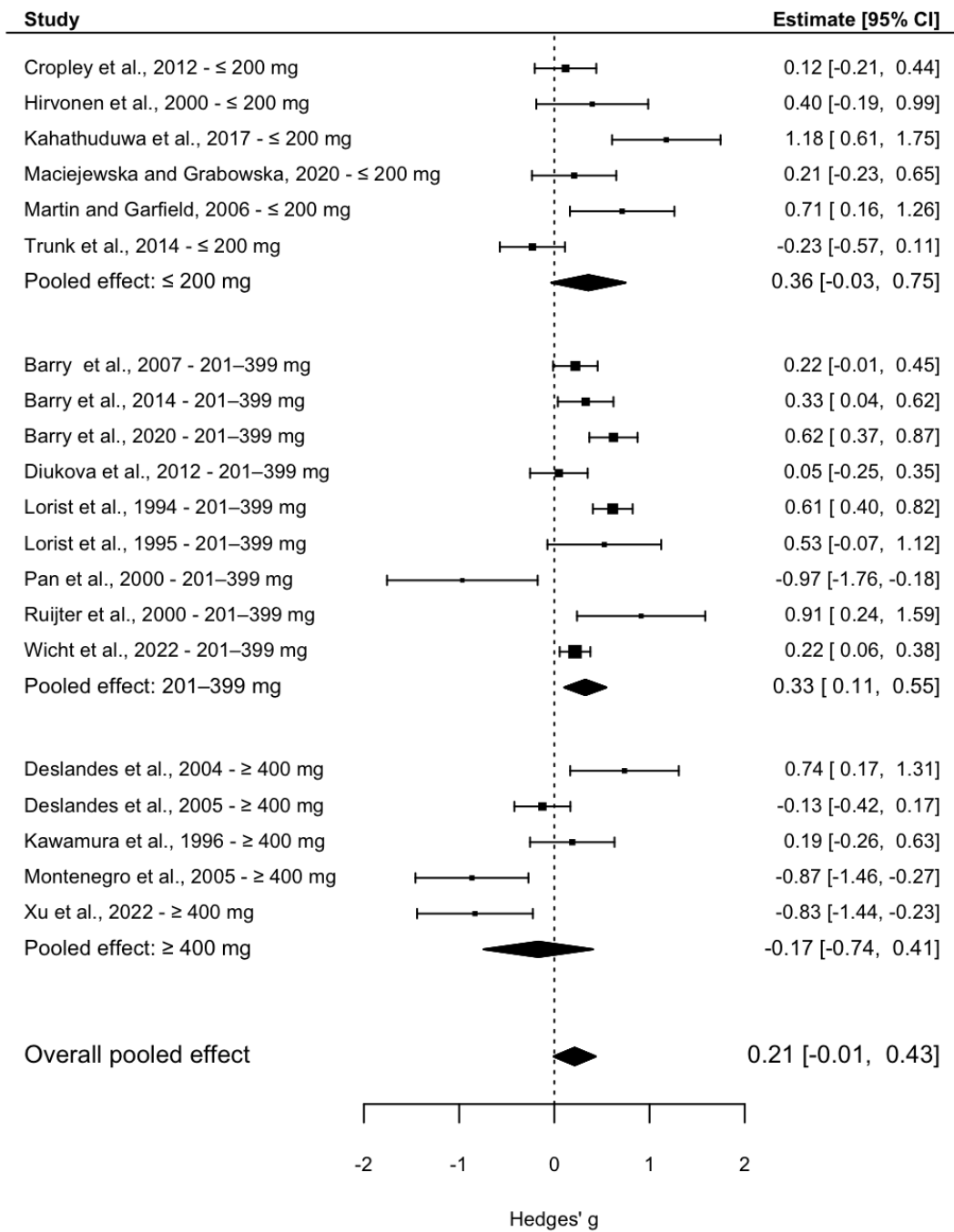


Figure 5

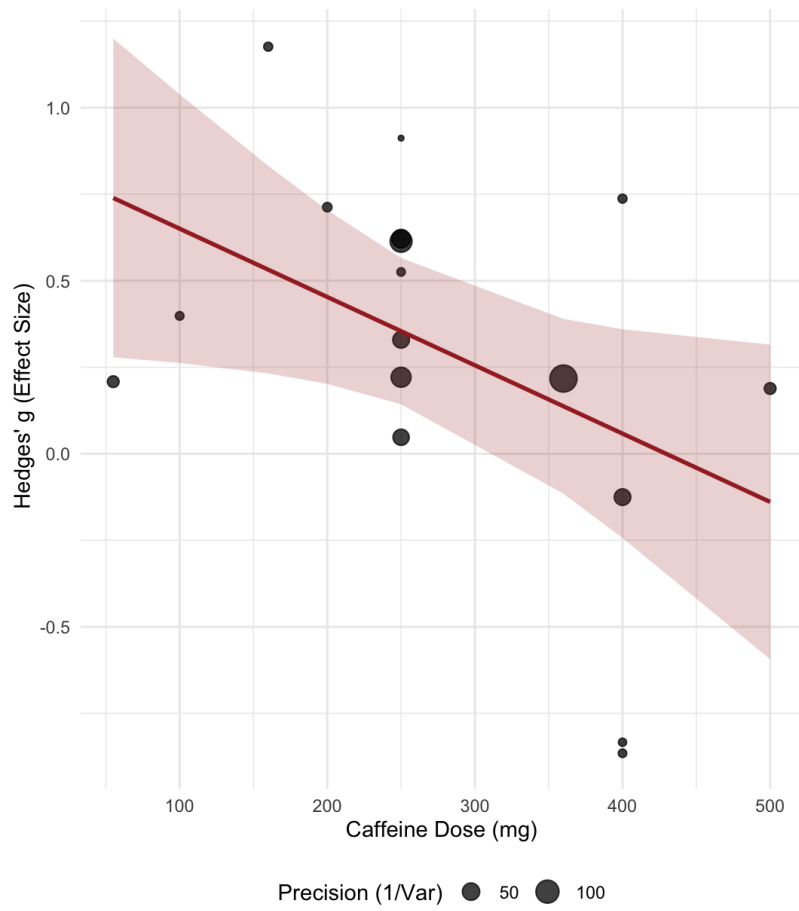
Meta-regression: effect size vs. caffeine dose

Table 1*Characteristics of the included studies*

Author s (year)	Qu alit y	Study desig n	Partici pant s (n femal e); M age	Dail y caff eine inta ke	Caffein e treatm ent	Caffein e dose	Time betwe n caffein e admini stration and EEG recordi ng	Experi mental task (modali ty; stimuli)	P300 nomen clature (time- windo w ms)	Regi on	Result s (P300 amplit ude and latenc y)
Barry et al. (2007) *	18	Pre- post	24 (16F); 22.9 y	2-4 “cup s”	Gelatin capsule	250 mg	30 min	Go/No Go (Auditor y; Tones)	P3a/P3 b; (270 - 400)	All	Ampli tude: Post- caff > pre- caff
Barry et al. (2014) *	17	Pre- post	24 (13F); 21.14 y	2-4 “cup s”	Gelatin capsule	250 mg	M = 35 min after caffeine M = 34.5 min after placebo	Go/No Go (Auditor y; Tones)	P3b; (250 - 350)	All	Ampli tude: Post- caff > pre- caff
Barry et al. (2020) *	16	Pre- post	24 (16F); 21.6 y	2-4 “cup s”	Gelatin capsule	250 mg	M = 36.4 min after caffeine M = 35.5 min after placebo	Go/No Go (Auditor y; Tones)	P3a/P3 b; (250 - 400)	All	Ampli tude: Post- caff > pre- caff
Chen et al. (2020)	21	Pre- post	16 (0F); 25.9 y	-	-	400 mg	Caffeine - 28th h EEG - 36th h	Go/No Go (Visual; Arrows)	P300; (300 - 500)	All	Ampli tude: Post- caff > pre- caff
Crople y et al. (2012) *	19	Pre- post	39 (19F); 62.5 y	1-8 “cup s”	Soluble decaffei nated coffee or Caffein ated coffee	5; 11; 167 mg	40 min	Oddball (Visual; Digits)	P3b; (280 - 400)	Post erior	Ampli tude: Post- caff > pre- caff
Debrah et al. (1996)	7	Pre- post	12 (7F); 33.16 y	<20 0 mg	-	250 mg	Not fixed	Oddball (Auditor y; Clicks)	P300; (-)	Post erior	Latenc y: ns
De Pauw et al. (2015)	21	Pre- post	10 (0F); 27 y	2 “cup s”	Mixture solution	0.3 g/25 ml	Immedi ately after caffeine	Stroop tasks (Visual; Letters)	P300; (200 - 400)	All	Ampli tude: ns Latenc y: ns
De Pauw et al. (2017)*	20	Pre- post	10 (0F); 22 y	0.5 ± 0.6 (<50 mg/ day)	Nasal spray	15 mg/ml	na	Stroop tasks (Visual; Letters)	P300; (240 - 470)	Ante rior	Ampli tude: Post- caff > pre- caff

Deslandes et al. (2004) *	19	Pre-post	15 (9F); 26 y	<4 "cups"	Gelatin capsule	400 mg	30 min	Oddball (Visual; Square and circle)	P300; (250 - 500)	All	Amplitude: Post-caff > pre-caff Latency: Post-caff < pre-caff
Deslandes et al. (2005) *	20	Pre-post	15 (9F); 26 y	1-2 "cups"	Gelatin capsule	400 mg	30 min	Oddball (Visual; Square and circle)	P300; (250 - 500)	All	Amplitude: Post-caff < pre-caff Latency: Post-caff < pre-caff
Deslandes et al. (2006) *	18	Pre-post	10 (5F); 30.4 y	1-2 "cups"	Capsule	400 mg	1 h	Oddball (Visual; Square and circle)	P300; (250 - 500)	Central	na
Díaz et al. (2022)	18	Between-groups	(EG)- 8 (3F); 29 y (CG)- 25 (12F); 23.9 y	-	-	.	Variable 1 h - 8 h	Oddball (Visual; Letters)	P300; (0 - 800)	All	Amplitude: (EG) > (CG)
Dimpfle et al. (2007) *	16	Pre-post	12 (0F); 45 y	-	Decaffeinated green tea extract	0.01% (Soft drink-500ml)	1 h - 4 h	Oddball (Visual; X and O)	P300; (-)	Central	Amplitude: Post-caff > pre-caff Latency: Post-caff < pre-caff
Diukova et al. (2012) *	25	Pre-post	14 (0F); 25.7 y	<36 mg	Gelatin capsule	250 mg	30 min	Oddball (Auditory; Sounds)	P300; (300 - 550)	All	Amplitude: Post-caff > pre-caff
Dixit et al. (2006) *	15	Pre-post	40 (0F)	-	Caffeine mixed with sugar, milk and 100 ml of water	3 mg/kg	40 min	Oddball (Auditory; Sounds)	P300; (-)	All	Amplitude: Post-caff > pre-caff Latency: Post-caff < pre-caff
Hirvonen et al. (2000) *	15	Pre-post	12 (5F)	-	Pill	100 mg	~ 50 min	Oddball (Auditory; Tones)	P3b; (-)	Posterior	Amplitude: Post-caff > pre-caff Latency:

Kahathuduwa et al. (2017) *	17	Pre-post	20 (0F); 21.9 y	<4 "cups"	Caffeine and 150 ml of distilled water	160 mg	30 min	Oddball (Auditory; Tones)	P300; (250 - 350)	Central	Post-caff < pre-caff Amplitude: Post-caff > pre-caff Latency: Post-caff < pre-caff
Kawamura et al. (1996) *	13	Pre-post	10 (3F); 28 y	-	-	500 mg	30 min	Oddball (Auditory) Single tone (Auditory; Tones)	P300; (250 - 500)	Posterior	Amplitude: Post-caff > pre-caff Latency: ns
Kenemans and Lorist (1995)	17	Pre-post	16 (8F); 21.5 y	5.9 "cups"	Mixture solution	3 mg/kg	~ 45 min	Visual selective attention (Visual; Squares)	P3; (740-950)	Anterior	Latency: Post-caff < pre-caff
Kerr et al. (1993)	7	Pre-post	8 (3F); 20-33 y	-	Caffeine in cola	400 mg	Not fixed	Oddball (Auditory; Clicks)	P300	-	Latency: Post-caff > pre-caff
Lorist et al. (1994) *	17	Pre-post	30 (14F); 21.2 y	2-7 "cups"	-	200 mg (beginning) + 50 mg (maintenance)	~45 min	Stimulus degradation task (Visual; Digits)	P300; (400 - 500)	All	Amplitude: Post-caff > pre-caff
								Stimulus-response compatibility task (Visual; Digits)	P300; (350 - 400)	Anterior	Amplitude: Post-caff > pre-caff
								Time uncertainty task (Visual; Digits)	P300; (300 - 450)	All	Amplitude: Post-caff > pre-caff
Lorist et al. (1995) *	19	Pre-post	(Y) 15 (7F); 21.1 y (E) 15 (11F); 64.5 y	2-7 "cups"	Caffeine, sugar and milk in a cup	200 mg (beginning) + 50 mg (maintenance)	~ 90 min	Memory task (Visual; Letters)	P3b; (350 - 450) P3b; (300 - 800)	All	Amplitude: Post-caff > pre-caff Latency: Post-caff <

Lorist et al. (1996)	19	Pre-post	16 (8F); 21.5 y	5.9 "cup s"	Caffeine dissolved in a cup of decaffeinated coffee	3 mg/kg	45 min	Oddball (Visual; Letters)	P3b; (300-600)	All	pre-caff Latency: Post-caff < pre-caff
Lorist et Snel (1997) *	22	Pre-post	16 (8F); 21.8 y	4-5 "cup s"	Caffeine dissolved in a cup of decaffeinated coffee	3 mg/kg	~ 70 min	Eriksen flanker task version (Visual; Letters)	P3; (300 - 800)	Posterior	Latency: Post-caff < pre-caff
Maciejewska and Grabowska (2020) *	23	Between-groups	(EG)-19; 26.1 y (CG)-21; 26.1 y	-	One cup of water with dissolved one effervescent tablet of energy boost dietary supplement	55 mg	30 min and 90 min	Oddball (Visual; Letters and numbers) Oddball (Auditory; Tones)	P300; (350 - 550) P300; (250 - 450)	Central and posterior All	Amplitude: (EG) > (CG)
Martin and Garfield (2006)*	18	Pre-post	16 (F); 24 y	0-600 mg	Caffeine pill	200 mg	45 min	Choice RT task (Visual; Letters)	P300; (270 - 530)	Posterior	Amplitude: Post-caff > pre-caff Latency: ns
Montenegro et al. (2005) *	18	Pre-post	15 (8F)	-	Capsule	400 mg	30 min	Oddball (Visual; Square and circle)	P300; (250 - 550)	All	Amplitude: Post-caff < pre-caff Latency: Post-caff < pre-caff
Pan et al. (2000) *	13	Pre-post	9 (0F); 30.7 y	1-2 "cup s"	Caffeine dissolved in a cup	5 mg/kg (275-375 mg)	30 min	Oddball (Auditory; Tones)	P300; (250 - 400)	All	Amplitude: Post-caff < pre-caff Latency: Post-caff < pre-caff
Rao et al. (2005)	15	Pre-post	40 (26F); 18-30 y	-	Energizing drink	40 mg	~ 30 min	Oddball (Visual; Checkboard)	P3; (400-500)	All	Amplitude: Post-caff > pre-caff

Reeves et al. (1999) *	8	Pre-post	13 (9F); 18-48 y	>300 mg	Dose they usually consume	-	na	Oddball (Auditory; Tones)	P300; (-)	-	Latency: Post-caff < pre-caff
								Oddball (Visual; Black and-white checker board)			Amplitude: Pre-caff > post-caff **
											Latency: Post-caff > pre-caff
Ruijter and Snel (1999) *	20	Pre-post	20 (15F); 20.9 y	3-5 “cups”	Dissolved in a cup of decaffeinated coffee	1; 3; 7.5 mg/kg	45 min	Grating task (Visual; Squares)	P300; (400 - 500)	All	Amplitude: Post-caff > pre-caff
											Latency: Post-caff > pre-caff
Ruijter et al., (2000) *	22	Pre-post	11 (7F); 22.4 y	3-6 “cups”	Mixture solution	250 mg	40 min	Oddball (Visual; Color)	P3b; (300-700)	All	Amplitude: Post-caff > pre-caff
Ruijter et al. (2000) *	20	Pre-post	12 (8F); 22.4 y	3-6 “cups”	Dissolved in a cup of normally brewed decaffeinated coffee	250 mg	~ 40 min	Grating task (Visual; White dots)	P300; (520 - 616)	All	Amplitude: Post-caff > pre-caff
Saville et al. (2018) *	22	Pre-post	15 (F); 21.6 y	129 ± 104 mg	Capsule	6 mg/kg	~ 2 h	Oddball (Visual; Coloured circles)	P3b; (250 - 750)	All	Amplitude: Post-caff < pre-caff
											Latency: Post-caff < pre-caff
Tieges et al. (2004) *	22	Pre-post	15 (7F); 20.4 y	1.51 - 5.38 “cups”	Dissolved in a cup of decaffeinated coffee	3 e 5 mg/kg	45 min	Task switching (Visual; Letters red and blue)	P300; (300 - 600)	All	Amplitude: Post-caff > pre-caff
Tieges et al. (2007) *	22	Pre-post	18 (9F); 21.6 y	2.7-8.6 “cups”	Dissolved in a cup of decaffeinated coffee	3 e 6 mg/kg	~ 40 min	Task switching (Visual; Letters)	P300; (300 - 600)	All	Amplitude: Post-caff > pre-caff

					nated coffee		45 min	red and blue)		pre- caff
Trunk et al. (2014) *	17	Pre- post	25 (9F); 21.07 y	-	Gelatin capsule	5; 10; 20; 100 mg (3mg/kg g)		Oddball (Visual; Square and circle)	P300; (400 - 700)	Post erior Amplit ude: Post- caff < pre- caff Latenc y: Post- caff < pre- caff
Vorkap ic (2009)	11	Betw een- grou ps	54 (F); 20.05 y	-	-	-	na	Oddball (Visual)	P300; (300 - 600)	All na
							20 min	Memor y task (Visual; Digits)	P300; (208 - 277)	Ampli tude: Post- caff > pre- caff Latenc y: Post- caff > pre- caff
Wicht et al. (2022) *	23	Pre- post	38; 23.15 y	1-4 "cup s"	Brewed coffee	240 + 120 mg				All Ampli tude: Post- caff > pre- caff Latenc y: Post- caff < pre- caff
								Go/No Go (Visual; White letters)	P300; (300 - 500)	
							4 h			Ampli tude: Post- caff < pre- caff Latenc y: Post- caff > pre- caff
Xu et al. (2022) *	18	Pre- post	14 (0F); 23.31 y	-	Gelatin capsule	400 mg		2-back task (Visual; Letters)	P300; (300 - 400)	All Latenc y: Post- caff > pre- caff

Note. Caff = Caffeine; CG = Control Group; E = Elderly; EG = Experimental Group; F = Female; h = hour; min = minutes; na = not available; nr = not reported; ns = non-significant findings; Y = younger group; y = years
 * = Studies that were included in the meta-analysis; ** = This result has the same meaning as post-caff > pre-caff, the difference is that in this study the participants were analysed on the habitual caffeine intake and then on caffeine abstinence.

Table 2*Statistics for caffeine effects on the P300 component*

Moderator	Category	k	Heterogeneity statistics		Hed ges' g	Confidence Interval - 95%		p
			Q	p		Lower limit	Upper limit	
Region	All	15	62.15	<.001	83 %	-0.07	0.51	.134
	Anterior	5	11.65	.020	67 %	0.11	0.82	.111
	Central	9	22.63	.004	68 %	0.17	0.61	<.001
	Posterior	11	28.87	.001	67 %	0.04	0.47	.019
Cognitive function	Attention	19	110.14	<.001	87 %	0.06	0.54	.011
	Executive	8	24.03	.001	80 %	0.03	0.57	.031
Stimuli	Auditory	11	49.18	<.001	85 %	0.13	0.67	.004
	Visual	17	78.98	<.001	86 %	0.00	0.47	.046
Dose	≤ 200 mg	6	21.29	<.001	78 %	-0.03	0.75	.072
	201-399 mg	9	31.76	<.001	99 %	0.11	0.55	.004

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		Heterogeneity statistics				Confidence Interval - 95%			
Moderator	Category	<i>k</i>	<i>Q</i>	<i>p</i>	<i>I</i> ²	Hed ges' g	Lower limit	Upp er limit	<i>p</i>
	≥ 400 mg	5	21.88	<.001	86%	-0.17	-0.74	0.41	.569
Overall		28	136.86	<.001	86%	0.30	0.12	0.48	.001

Table 3*Meta-regression analysis statistics*

Covariate	<i>k</i>	Coefficient	Confidence Interval		<i>p</i>
			– 95%		
			Lower Limit	Upper Limit	
Percentage of females	27	0.00	-0.00	0.01	.286
Age	24	-0.01	-0.03	0.01	.265
Dose	17	-0.00	-0.00	-0.00	.036
Time between caffeine administration and EEG recording	24	-0.00	-0.01	0.00	.052

Highlights

- Caffeine shows an inverted U-shaped dose–response effect on P300 amplitude.
- Moderate caffeine intake (201–399 mg) is associated with increased P300 amplitude.
- No significant differences in P300 latency were observed after caffeine intake.